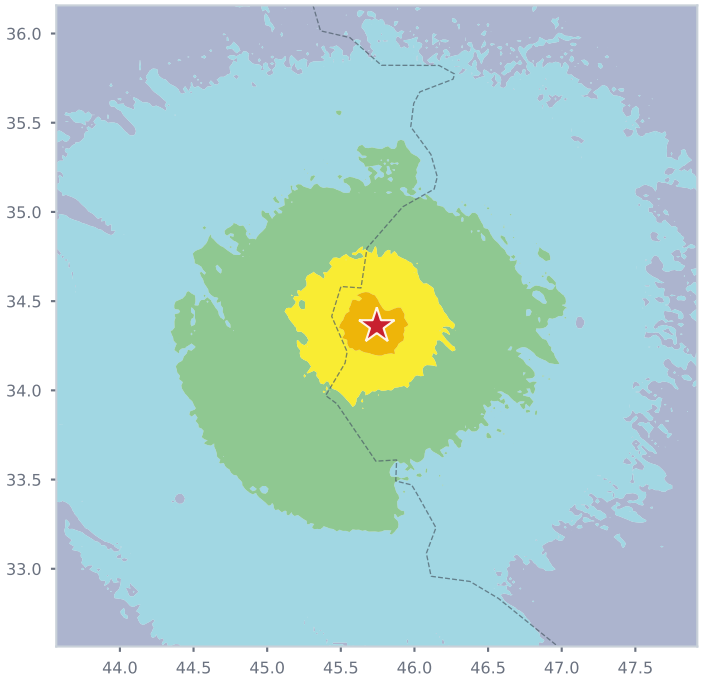
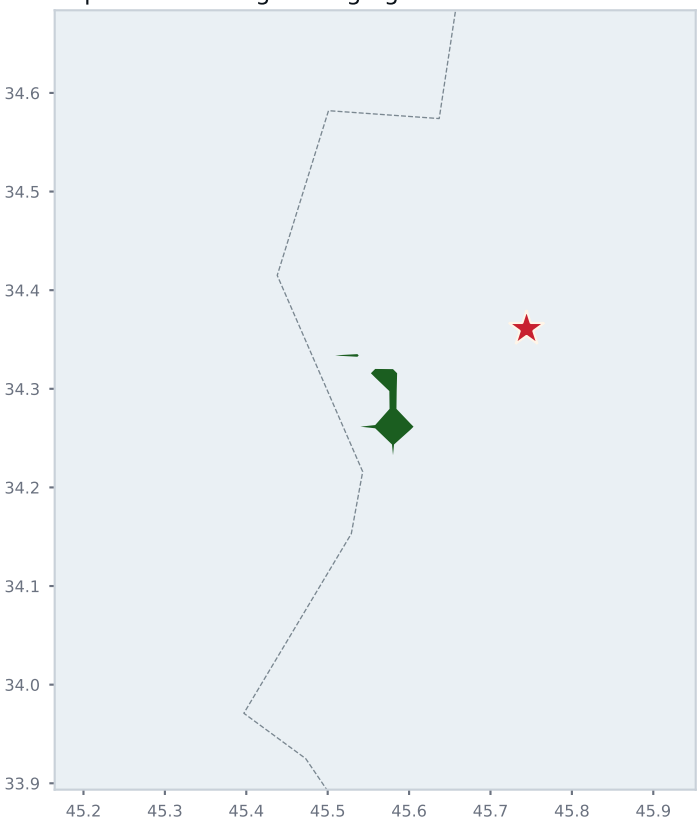
Shaking intensity (IMS-25)




Expected building damage grade




IMS-25 intensity




Expected damage grade (IMS-25)




Damage alert

**ORANGE** Widespread damage; many buildings unusable **Provisional** **Transit occupancy**

Buildings heavily damaged (DG3 or worse)

## 1.2 thousand to 13 thousand

Central estimate 3.7 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

## 5.3 million

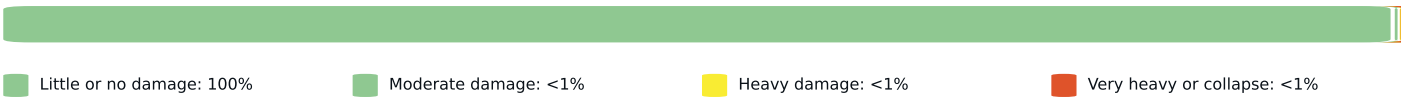
21% of them inferred from built-up area, not observed footprints.

People by shaking level

| | |
|---|---|
| **VII** | 190 |
| **VI** | 285 thousand |
| **V** | 1.9 million |
| **IV** | 16 million |

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage



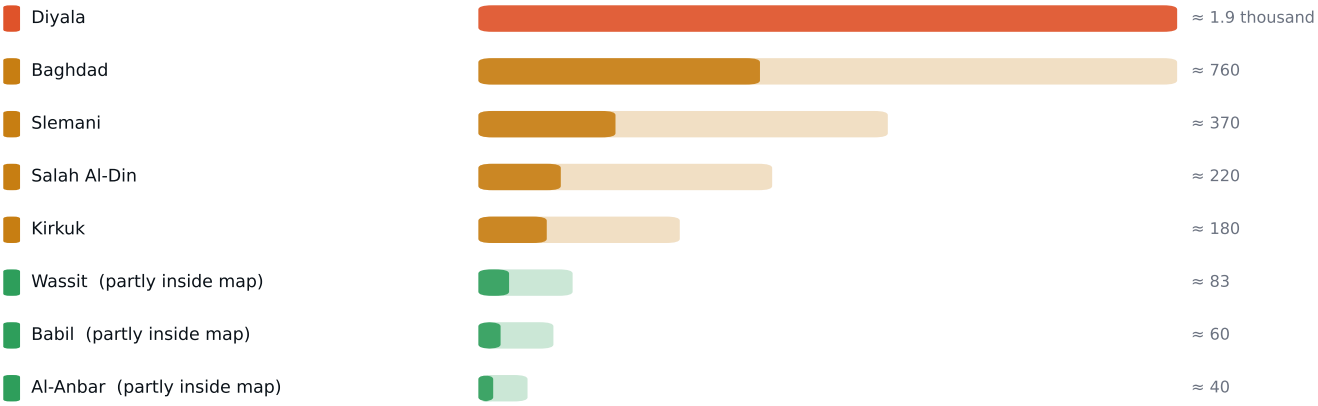
Little or no damage: 100% | Moderate damage: <1% | Heavy damage: <1% | Very heavy or collapse: <1%

## Which buildings failed

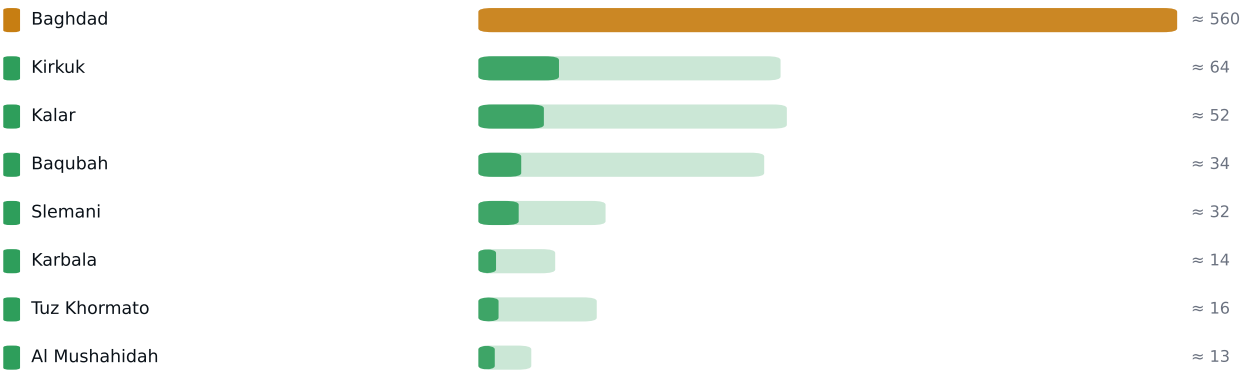
Share of each type's own buildings heavily damaged, worst first

| | | | |
|---|---|---|---|
| Rubble stone, fieldstone | class A | **0.3%** | ≈ 440 buildings |
| Adobe, earth brick | class A | **0.2%** | ≈ 320 buildings |
| Manufactured stone units, timber floors | class B | **0.1%** | ≈ 510 buildings |
| Simple stone | class B | **0.1%** | ≈ 360 buildings |
| RC flat slab and others | class B | **0.0%** | ≈ 2 buildings |
| Manufactured stone with RC floors | class C | **0.0%** | ≈ 310 buildings |
| RC frame — no / low code (ERD-L) | class C | **0.0%** | ≈ 59 buildings |

## Governorates most affected: heavily damaged buildings, central estimate and 90% range




## Cities most affected



## Every governorate inside the map

| Area | Buildings | Heavy (P50) | 90% range | % of its stock |
|---|---|---|---|---|
| Diyala | 602 thousand | 1.0 thousand | 510 to 7.0 thousand | 0.17% |
| Baghdad | 2.1 million | 470 | 290 to 2.6 thousand | 0.02% |
| Slemani | 588 thousand | 220 | 120 to 1.1 thousand | 0.04% |
| Salah Al-Din | 411 thousand | 140 | 87 to 800 | 0.03% |
| Kirkuk | 549 thousand | 120 | 70 to 540 | 0.02% |
| Wassit | 195 thousand | 52 | 28 to 260 | 0.03% |
| Babil | 219 thousand | 40 | 20 to 200 | 0.02% |
| Al-Anbar | 172 thousand | 29 | 17 to 130 | 0.02% |
| Kerbala | 201 thousand | 21 | 11 to 97 | 0.01% |
| Hawler | 255 thousand | 20 | 11 to 78 | 0.01% |
| Ninewa | 720 | none | none to none | 0% |
| Maysan | 660 | none | none to none | 0% |

## How this report was made

**Hazard engine** Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

**Observations** 0 seismic stations, 122 felt-report cells (felt channel: dyfi)

**Damage scale** IMS-25

### Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

### What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml20180002, version 2, generated 19 Oct 2026, 13:36 UTC.